RESOUND: Revealing Sound Driven Avian Barriers in Urbanized Landscapes

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MOTIVATION:

- Songbird populations have experienced consistent long-term decline due to habitat loss, reduced survival, and environmental stressors.
- Anthropogenic noise is an often overlooked contributor, masking communication signals and increasing ecological vulnerability.
- Even environments with low-to-moderate overall loudness can produce ecological impacts comparable to highly urbanized regions.
- However, there remains a logical and computational gap between measuring noise and quantifying its ecological consequences.
- How can we model sound data in a way that supports data-driven conservation and urban planning?

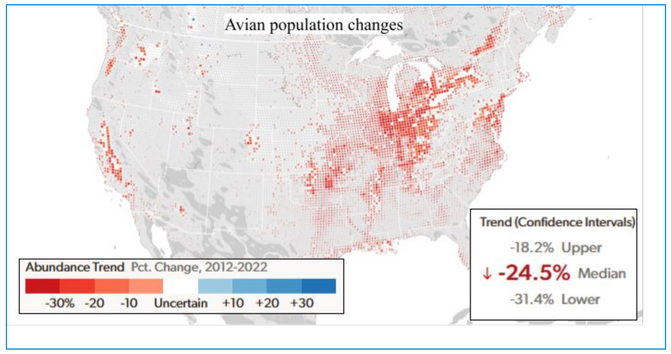

Fig. 1: eBird-derived visualization of long-term avian population trends across the United States, illustrating widespread declines in bird abundance over time.(Credit: Cornell Ebird)

GOALS AND OBJECTIVES:

- Translate urban sound data into interpretable ecological impact metrics.
- Incorporate sound source composition and temporal persistence beyond average loudness.
- Generate Avian Resistance Maps (ARMs) to visualize spatial acoustic constraints.
- Compare resistance patterns with independent bird observation data.
- Support scalable, data-driven conservation strategies.

EXPERIMENTAL DESIGN:

- **Urban Dataset Analysis (Washington Square Park)**
  - Trained a CNN using annotated SONYC urban audio data
  - Applied the trained model to held-out SONYC recordings
  - Generated Avian Resistance Maps (ARMs) for Washington Square Park and surrounding urban blocks within the SONYC dataset
- **Pilot Field Deployment (DuPage Riverfront Park)**
  - Deployed six AudioMoth acoustic sensors at spatially distributed locations as shown in Figure 3
  - Collected passive environmental audio recordings
  - Processed recordings through the RESOUND pipeline to generate local ARM results

Figure 3: AudioMoth sensor deployment used for pilot acoustic data collection

COMPONENTS AND PHASES OF THE RESOUND WORKFLOW:

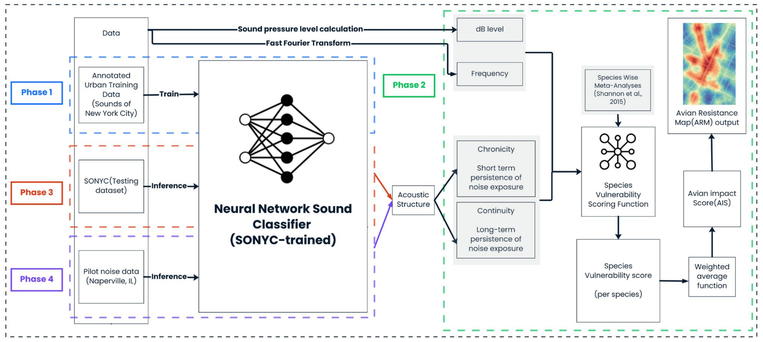

Figure 2: Overview of the RESOUND framework

- **Phase 1: Urban Sound Classification**
  - Train a convolutional neural network (CNN) on the SONYC dataset to identify dominant anthropogenic sound sources (e.g., traffic, construction, mechanical noise).
- **Phase 2: Acoustic & Temporal Feature Extraction**
  - Extract Sound intensity, Frequency characteristics, Chronicity (short-term persistence) and Continuity (multi-day recurrence)
- **Phase 3: Avian Impact Scoring**
  - Combine acoustic features into an Avian Impact Score (AIS) that represents relative acoustic constraint at each location.
- **Phase 4: Spatial Mapping & Validation**
  - Aggregate AIS spatially to generate Avian Resistance Maps (ARMs) and compare results with independent bird observation data (eBird).

REFERENCES

1. Aletta, F., De Coensel, B., & Lindborg, P. (2021). Editorial: Human Perception of Environmental Sounds. Frontiers in Psychology, 12, 714591. https://doi.org/10.3389/fpsyg.2021.714591
2. Barber, J. R., Crooks, K. R., & Fristrup, K. M. (2010). The costs of chronic noise exposure for terrestrial organisms. Trends in Ecology & Evolution, 25(3), 180–189. https://doi.org/10.1016/j.tree.2009.08.002
3. Fink, D., Auer, T., Johnston, A., Strimas-Mackey, M., Ligocki, S., Robinson, O., Hochachka, W., Jaromczyk, L., Crowley, C., Dunham, K., Stillman, A., Davies, I., Rodewald, A., Ruiz-Gutierrez, V., & Wood, C. (2023). EBird Status and Trends, Data Version 2022 [Dataset]. https://doi.org/10.2173/ebirdst.2022
4. Halfwerk, W., Holleman, L. J., Lessells, C. M., & Slabbekoorn, H. (2011). Negative impact of traffic noise on avian reproductive success. Journal of Applied Ecology, 48(1), 210–219. https://doi.org/10.1111/j.1365-2664.2010.01914.x
5. Meillère, A., Buchanan, K. L., Eastwood, J. R., & Mariette, M. M. (2024). Pre- and Postnatal Noise Directly Impairs Avian Development, With Fitness Consequences. Science, 384(6694), 475–480. https://doi.org/10.1126/science.ade5868
6. Pijanowski, B. C., Villanueva-Rivera, L. J., Dumyahn, S. L., Farina, A., Krause, B. L., Napoletano, B. M., Gage, S. H., & Pieretti, N. (2011). Soundscape ecology: The science of sound in the landscape. BioScience, 61(3), 203–216. https://doi.org/10.1525/bio.2011.61.3.6
7. Salamon, J., & Bello, J. P. (2017). Deep convolutional neural networks and data augmentation for environmental sound classification. IEEE Signal Processing Letters, 24(3), 279–283. https://doi.org/10.1109/LSP.2017.2657381
8. Salamon, J., Bello, J., Silva, C., Nov, O., DuBois, R. L., Arora, A., Mydlarz, C., & Doraiswamy, H. (2018). SONYC: A system for the monitoring, analysis, and mitigation of urban noise pollution. Communications of the ACM, 61(10), 100-108. https://doi.org/10.48550/arXiv.1805.00889
9. Slabbekoorn, H., & Ripmeester, E. A. P. (2008). Birdsong and anthropogenic noise: Implications and applications for conservation. Molecular Ecology, 17(1), 72–83. https://doi.org/10.1111/j.1365-294X.2007.03487.x
10. Sullivan, B. L., Aycrigg, J. L., Barry, J. H., et al. (2014). The eBird enterprise: An integrated approach to development and application of citizen science. Biological Conservation, 169, 31–40. https://doi.org/10.1016/j.biocon.2013.11.003
11. Ware, H. E., McClure, C. J. W., Carlisle, J. D., & Barber, J. R. (2015). A Phantom Road Experiment Reveals Traffic Noise Is an Invisible Source of Habitat Degradation. Proceedings of the National Academy of Sciences, 112(39), 12105–12109. https://doi.org/10.1073/pnas.1504710112

RESULTS AND VALIDATION:

Urban Case Study: Washington Square Park

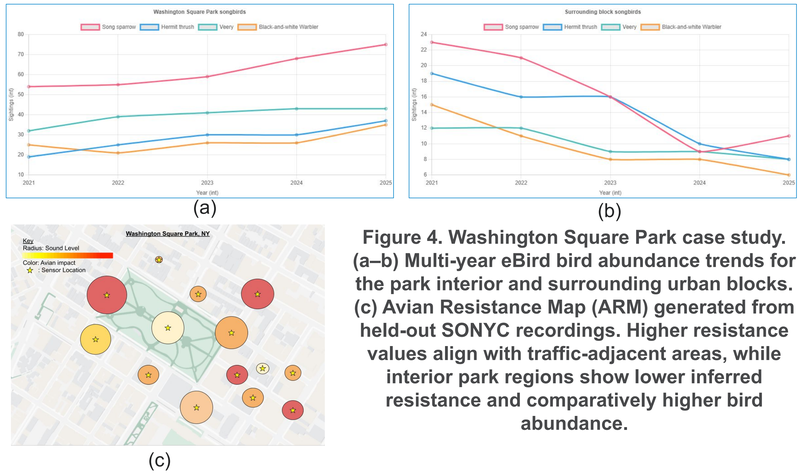

Figure 4. Washington Square Park case study. (a–b) Multi-year eBird bird abundance trends for the park interior and surrounding urban blocks. (c) Avian Resistance Map (ARM) generated from held-out SONYC recordings. Higher resistance values align with traffic-adjacent areas, while interior park regions show lower inferred resistance and comparatively higher bird abundance.

- The ARM reveals spatial variation in inferred acoustic resistance across the park and surrounding urban blocks.
- Although sound intensity remains high within parts of the park, resistance values are lower when dominant sources are speech and transient activity rather than persistent traffic or construction noise.
- Multi-year eBird observations show relatively higher bird abundance within the park interior and reduced abundance along traffic adjacent edges.
- These findings suggest that sound source composition and temporal persistence influence ecological impact beyond decibel level alone.

Pilot Field Study: DuPage Riverfront Park

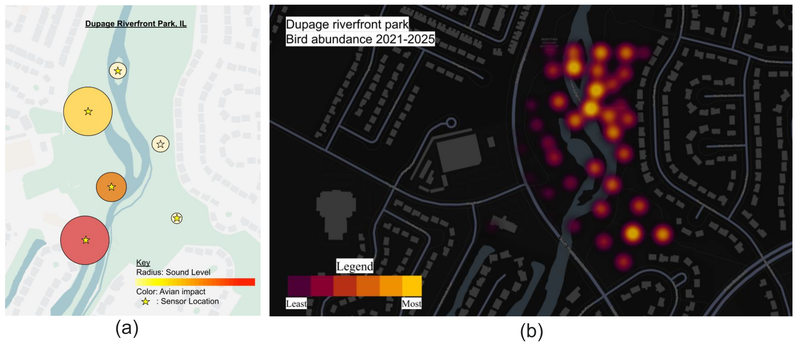

Figure 5. DuPage Riverfront Park pilot study. (a) Avian Resistance Map generated from six AudioMoth sensor locations. (b) eBird bird abundance heatmap for the same region. Lower inferred resistance corresponds to areas of higher observed bird abundance.

- Six AudioMoth sensors were deployed across spatially distributed locations within the park.
- The generated ARM shows variation in inferred resistance along the river corridor.
- Regions with lower resistance correspond to areas of higher bird observation density in eBird data.
- These preliminary results suggest that RESOUND captures meaningful acoustic variation in a suburban field setting.

CONCLUSIONS:

- RESOUND translates urban sound recordings into measurable indicators of ecological constraint. Results from both an urban case study and a suburban pilot deployment suggest that sound source composition and temporal persistence, not loudness alone, shape inferred avian resistance. This scalable framework supports more informed, data-driven urban conservation.